# INDEPENDENT, ALTERNATIVE, AND COMMUNITY MEDIA AN OPEN READER

Edited and with an introduction by
David W. Park & Meghan Grosse

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# Independent, Alternative, and Community Media: An Open Reader

**Edited by David W. Park & Meghan Grosse**

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CHAPTER 15

# From Revolution to Reform: Recommendations for Spectrum Policy in Transitional Tunisia

Moez Chakchouk, Danielle Kehl, Jochai Ben-Avie, and Kate Coyer

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*Abstract*

Will the revolutionary changes of the "Arab Spring" result in more open and democratic media in the affected countries in the Middle East and North Africa? Using developments in Tunisia as a case study, the authors argue that democratically reformed media, Internet, and spectrum policies can lead to a more enabling and pluralistic environment that encourages new market entrants. Progress has been made, but ongoing challenges in structural and institutional reforms remain. The authors make a series of policy recommendations in the hope that Tunisia can be a model for its regional and global peers.

## *Introduction*

In December 2010, Tunisian fruit vendor Mohamed Bouazizi set himself on fire to protest the pervasive corrupt practices of his municipal government. His actions sparked widespread demonstrations in Tunisia that were magnified by both traditional news outlets and social media. These protests, in turn, triggered a wave of social unrest around the Middle East

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in what would come to be called the Arab Spring. In January 2011, after 24 years in power, Tunisian President Zine el-Abidine Ben Ali became the first Arab leader to step down in response to the pressure against his rule. The following October, Tunisia held the first post-Arab Spring election, and the country is now in transition towards a more robust democracy through constitutional reform, a process that involves overhauling many of the policies and regulatory regimes over which Ben Ali once maintained a tight grip.

As a nation, Tunisia has long stood apart from others in the region, ranking first among North African countries in the 2010–2011 World Economic Forum Global Competitiveness index [1] and boasting a stronger economy, a better-educated population, and relatively low poverty levels compared to many of its neighbors. However, the Ben Ali government also kept a notoriously close watch over its citizens, maintaining tight control over the media environment and restricting freedom of expression. While Tunisia has long had uncharacteristically high levels of mobile and broadband penetration when compared to the rest of the Middle East and North Africa (a reflection, at least in part, on the strength of its economy and overall competitiveness), its citizens were not able to enjoy many of the corresponding benefits of ubiquitous communications or an open Internet. Up until the day he fled the country, Ben Ali exercised significant institutional and personal control over the Internet, broadcasting, and the telecommunications sector, resulting in pervasive censorship and an official regulatory structure that was largely superfluous, lacking both sophistication and clarity. Today, foreign investors and independent broadcasters continue to find it difficult to navigate the system successfully, and many of the old rules and processes have yet to be redressed. The interim government has recognized the need to increase access to traditional and web-based media and encourage freedom of expression, and thus made the reform of media and telecommunications structures a key piece of their platform. However, a truly enabling and pluralistic environment that encourages new market entrants has yet to emerge.

As Tunisia continues its transition to democracy, civil servants and other political actors are facing several competing priorities, of which regulatory reform to ensure the free flow of information and a more democratic media environment is just one. Free expression, access to information, and freedom of the media in Tunisia must be backed by regulatory institutions and policies that encourage and enable media pluralism, access for new players (private, public, and community) to enter the market, and independent regulatory systems and media outlets. Spectrum regulation, which determines the frequencies that are assigned to broadcasters and other wireless technologies, is critical to this process. Spectrum policy designed to enhance democracy can help Tunisia maintain its competitive edge through increased broadband Internet penetration, widespread sharing of information and exchange of ideas, technological innovation, and long-term economic growth. [2] Spectrum policy should therefore be a major priority in the restructuring of Tunisia's media environment, particularly because of its impact on wireless communications and Internet connectivity.

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[1] Klaus Schwab, ed., "The Global Competitiveness Report 2010–2011," report, World Economic Forum (2011), accessed Dec. 13, 2013, http://www3.weforum.org/docs/WEF_GCR_Report_2011-12.pdf. Tunisia ranked 32nd overall on the index.

[2] Turgut Cankorel and Lara Aryani, "A Spectrum of Regulations: Mobile Telecom Regulation in the Middle East and North Africa," *Convergence* 5 (2009): 165–175.

The public commitment to overhauling government policies to enable a freer media environment creates a unique opportunity to explore different spectrum policies and leverage these regulatory reforms for democratic and developmental gains.[3] This process, especially if it adheres to principles of access and openness, could serve as an example for other countries in the region that are seeking to reform communications policies to similar ends.

This article will begin by discussing the media environment, broadcasting and telecommunications regulations, and spectrum allocation policies in Tunisia before the January 2011 revolution. After identifying areas of deficiency in the old system, the article will summarize the progress that has been made in the post-revolutionary period, and address ongoing challenges and potential opportunities. One of the key objectives will be to reform existing structures and institutions as a part of Tunisia's overall shift toward democracy and new attitudes toward media and access to communications.

## *Broadcasting and Telecommunications in Pre-Revolutionary Tunisia*

Under former President Zine el-Abidine Ben Ali, the majority of institutions and key players in the Tunisian telecommunications and media sector were tied either directly or indirectly to the government, which had a hand in everything from the development of infrastructure to licensing and content regulation. Independent media groups, rather than enjoying autonomy and operating under a clear and transparent set of guidelines, were obliged to survive as best they could and find ways to coexist under the uncertainties of the Ben Ali regime.[4] As such, there was a disconnect in the Tunisian telecommunications and broadcasting landscape: Tunisia had some of the most robust infrastructure and highest rates of both mobile and Internet penetration in the Middle East/North Africa (MENA) region, but its government restricted individual freedom of expression and established a regulatory framework that gave the state enormous control—even in comparison to other MENA countries.

In terms of infrastructure development, Tunisia has long outpaced most of its neighbors, benefiting from a relatively stable economy, lower levels of poverty, and a citizenry more equipped to afford both the equipment and the associated operational costs of communications and broadcast networks.[5] As a result of this buildout, Tunisian citizens have near-universal access to mobile phone service and some of the lowest broadband prices in Africa. In 2008, Tunisia reported approximately nine million mobile subscribers out of a population of 10.2 million, and that number continued to rise until the number of mobile subscriptions surpassed the number of citizens about a year later.[6] Tunisia also reported 1.7 million Internet users

[3] Sana Ajmi, "Media Reform in Tunisia: A Long Way to Go?" *Tunisia Live*, Aug. 16, 2012, accessed Dec. 13, 2013, http://www.tunisia-live.net/recovery/media-reform-in-tunisia-a-long-way-to-go/.

[4] Joan Barata Mir, "The New Tunisian Legislative Framework: A Focus on Press and Audiovisual Media," white paper, Internews, Feb. 2012, accessed Dec. 14, 2013, http://www.internews.org/sites/default/files/resources/Internews_TunisiaMediaLaw_2012-02.pdf.

[5] The Davos Forum has ranked Tunisia first in Africa in terms of Networked Readiness every year since 2006. Ministry of Information and Telecommunications (Republic of Tunisia), "Classement de la Tunisie selon le Forum Economique Mundial de Davos Relatif Aux TIC durant la periode 2006-2012," report (2013), accessed Dec. 13, 2013, http://www.mincom.tn/fileadmin/PDF/Indicateurs_TIC/Positionnement_Tunisie_2006_2009/Positionnement-Tunisie-NRI_2006-2012-Fr.pdf.

[6] Open Net Initiative, "Internet Filtering in Tunisia," report, Aug. 7, 2009, accessed Dec. 13, 2013, https://opennet.net/sites/opennet.net/files/ONI_Tunisia_2009.pdf.

in 2008, and by March 2010, the number of Internet users had nearly doubled, reaching a reported 3.6 million. At roughly 33.9% of the population, the 2010 figure was significantly higher than the African average of 11.4% and well above the 31.7% mean for the Arabic Middle East at that time.[7] Yet despite these positive statistics, the regulatory framework in Tunisia remained stagnant in the pre-revolutionary period, which stifled independent broadcasting of radio and television. It also prevented regulators from responding to the demands created by the explosive growth of wired and wireless access.

## *Overview of Broadcasting and Telecommunications Regulation before the Revolution*

Historically, the Ministry of Information and Communication Technologies has overseen the broadcasting and telecommunications industries at the cabinet level in Tunisia. The Ministry is broadly tasked with strategic planning to grow the domestic ICT sector and attract more international investment. The agencies under the Ministry's umbrella, including the National Broadcasting Corporation (Office National de la Télédiffusion, ONT), the National Telecommunications Commission (Instance Nationale des Télécommunications, INT), the National Frequencies Agency (Agence Nationale des Fréquences, ANF), the National Digital Certification Agency (Agence Nationale de Certification Electronique, ANCE), the Center of Telecom Studies and Research (Centre d'Etudes et de Recherche des Télécommunications, CERT), and the National Agency for Computer Security (Agence Nationale de la Sécurité Informatique, ANSI), have together been responsible for regulating broadcasting, telecommunications, and data and digital confidence. The role of Internet regulation fell originally to the Tunisian Internet Agency, a semi-private company with strong ties to the regime,

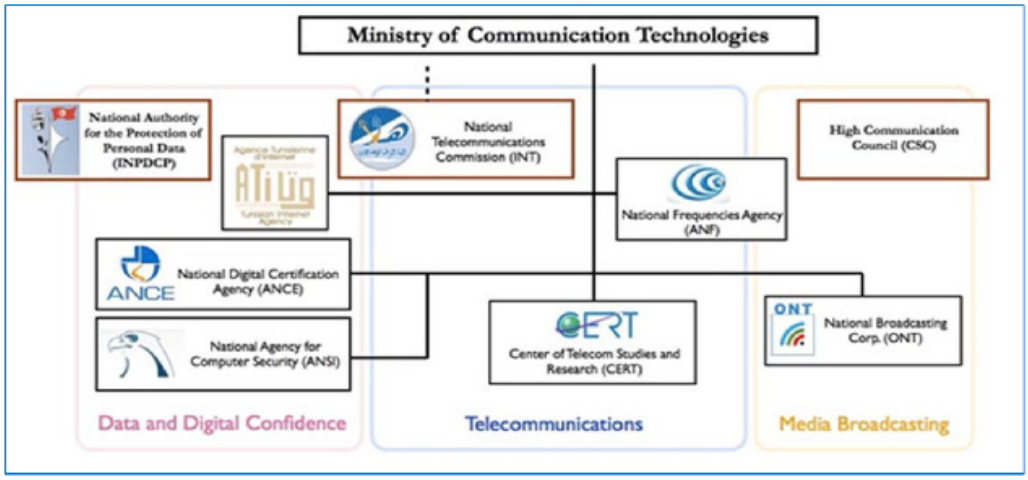



Figure 1: Regulatory agencies and authorities involved with the Ministry of Communication Technologies in regulation prior to the Tunisian revolution.

although later some of its authority was transferred to the National Telecommunications Commission. Figure 1 below illustrates the complexity of the regulatory system in Tunisia

[7] Data compiled from Internet World Stats, accessed Dec. 13, 2014, http://www.internetworldstats.com/stats1.htm#africa.

prior to the 2011 revolution, with nine different agencies and authorities overseen under the umbrella of the Ministry of Communication Technologies.

The public broadcasting landscape consisted of two television channels, Tunis 7 and Tunis 21 (now Wataniya 1 and Wataniya 2), and nine radio stations, four of which were national and five regional. The process for granting licenses to private media (including Hannibal-TV, Nessma-TV Mosaique-FM, Jawhra-FM, Shems-FM, and Express-FM) was largely opaque, which created a strong barrier to the development of the broadcasting sector. All licenses had to be authorized by the Minister of Information. Approval was further contingent upon a favorable advisory from an independent review board—the High Communications Council (Conseil Supérieur de la Communication, CSC), which was created by Decree No. 238 on January 30, 1989 and reorganized in 2008 by Law No. 2008-30 in preparation for a plan to liberalize the audiovisual sector. In reality, the review board was strongly tied to the regime and Ben Ali's advisers in the Presidential Palace, and most licenses were granted to government allies. Researchers at the Committee to Protect Journalists found that since the government started accepting applications in 2003, licenses had only been granted to friends and family of Ben Ali. For example, Cyrine Ben Ali, the President's daughter, owned Radio Shems FM, while Radio Zitouna for the Holy Koran belonged to Sakher El Matri, his son-in-law.[8]

Spectrum allocation was done through the National Frequencies Agency. The ANF allocated broadcasting bands to the ONT, the only entity with the right to own broadcasting transmitters. The ONT then distributed frequency bands to authorized radio and TV channels. As a result, private broadcasters in Tunisia had to first receive authorization from the government and then sign an agreement with the ONT to obtain a license to operate on a specific frequency.

The dynamics were similar in the telecommunications sector, where there was a largely state-controlled infrastructure and a vague regulatory framework. The largest provider of wired and mobile services in the country is Tunisie Telecom, which was established by the state in 1995. Although the government opened the mobile market to more competition in 2002, it did little to encourage real competition or create a climate conducive to foreign investment.[9] Consequently, although the government promoted access to mobile telephone and Internet services for its citizens, the regulatory framework did not reflect these policies.

There is perhaps no better example of the lack of structural institutions in pre-revolutionary Tunisia than the regulation of the Internet, which was made available throughout the country via terrestrial, satellite, and mobile data service providers but ultimately controlled by a single entity. In 1996, the government created the Tunisian Internet Agency *(the Agence Tunisienne d'Internet, ATI)* to serve as the central Internet authority, regulating the country's IP addressing and DNS services and operating the unique node for IP interconnection to the global Internet. When it reassigned all web services from the Institut Régional des Sciences Informatiques et des Télécommunications *(IRSIT, the research institute that was responsible for developing the Internet in Tunisia and connecting it to the wider world)* to the ATI, the government gave the ATI an unprecedented amount of responsibility and control over the network.

[8] Committee to Protect Journalists, "Attacks on the Press 2010: Tunisia," Feb. 15, 2011, accessed Dec. 13, 2013, http://www.cpj.org/2011/02/attacks-on-the-press-2010-tunisia.php.

[9] Moez Chakchouk, Presentation at the 3rd Arab Bloggers Meeting, Oct. 3–6, 2011, accessed Dec. 20, 2012, http://www.slideshare.net/mchakchouk/ati-ab1103102011-9547841

Consumer services were contracted out to Planet Tunisie ISP, the country's first commercial ISP, which was also owned by Cyrine Ben Ali. The government authorized additional ISPs in 1997 and the early 2000s, but they had to lease fixed-line access from Tunisie Telecom and international bandwidth from the ATI rather than controlling their own infrastructure and international connectivity. [10]

Under the Tunisia Telecommunications Code of 2001, the government established the National Telecommunications Commission (Instance Nationale des Telecommunications, INT) to serve as the lead regulatory agency in the telecommunications sector. Under the 2008 revisions of the Telecommunications Code, the INT was given an even broader mandate, ostensibly becoming an independent regulator of the telecommunications market with complete financial autonomy. The regulation of Internet services, which had previously been overseen by the Ministry of Information and Communications Technologies via the ATI, was also brought under the INT's purview in the restructuring. As a result, the INT took control of domain name regulation in 2009 while the ATI retained the technical role as registrar. In practice, of course, the Ministry of ICT had a heavy influence on INT decisions before the revolution, so the transfer of responsibilities from the ATI to the INT did little to actually alter the oversight process.

While the Internet regulation put in place in March 1997 (when only a few thousand Tunisians had connections) remained largely unchanged in the 14 years that followed, the ATI's censorship regime evolved a great deal, growing in both intensity and sophistication in response to network usage and expansion. As more Tunisians came online, the ATI spent an increasing amount of time and resources engaging in censorship on a massive scale. The Agency's policies were designed to shroud the work of the government, minimizing transparency and giving Ben Ali the ability to use the media for self-promotion and propaganda while silencing his critics. In the decade leading up the revolution, the ATI built both the technical infrastructure and the human resources capacity to implement keyword filtering, DNS and URL blocking, distributed denial of service (DDoS) attacks on sites that criticized the political regime, profile and mail hacking of its citizens, and more. [11] In addition to filtering and monitoring, the government arrested cyber dissidents who made both online and offline statements denouncing Internet censorship, even detaining and torturing a number of web activists during the 2005 World Summit on the Information Society, which was hosted by Tunisia. [12]

Consequently, Tunisia—often portrayed as more modern and liberal than many of its neighbors in the MENA region—began to draw international attention due to its systematic censorship regime. In 2009, the Committee to Protect Journalists ranked Tunisia seventh

[10] Katherine Maher and Jillian C. York, "Origins of the Tunisian Internet," in *State Power 2.0: Authoritarian Entrenchment and Political Engagement Worldwide*, ed. Muzammil M. Hussain and Philip N. Howard (London: Ashgate, 2013), 19–32.

[11] Sami Ben Gharbia, "A First Glimpse at the Internet Filtering in Tunisia," Jillian C. York, trans., Global Voices Advocacy, Aug. 18, 2010, accessed Dec. 13, 2013, https://tunisistan.appspot.com/advocacy.globalvoicesonline.org/2010/08/18/a-first-glimpse-on-the-internet-filtering- in-tunisia/.

[12] Ben Wagner, "'I Have Understood You': The Co-Evolution of Expression and Control on the Internet, Television and Mobile Phones during the Jasmine Revolution in Tunisia," *International Journal of Communication* 5 (2011): 1296; "Tunisia: Internet Repression Casts Pall on Web Summit," Human Rights Watch, Nov. 15, 2005, accessed Dec. 13, 2013, http://www.hrw.org/en/news/2005/11/14/tunisia-internet-repression-casts-pall-web-summit.

on its list of the "Ten Worst Countries To Be a Blogger," placing it in the same league as nations like Burma, Iran, Syria, Cuba, Saudi Arabia, and Turkmenistan.[13] The following year, Tunisia made it onto the Reporters Without Borders "Enemies of the Internet" list, for having "wagered on a[n] infrastructure development strategy while keeping a tight control over the Web's political and social content."[14]

## *Changes in the Air: Satellite Broadcasting, Pirate Radio, and Cyber-Dissent*

The rise of satellite TV from the late 1990s onward began to shift the broadcasting landscape to a limited degree, as it enabled content producers to exploit the relative lack of regulation compared to traditional broadcasting media and take advantage of the technical challenges of blocking it. Satellite infrastructure would prove particularly problematic for the Ben Ali regime during the Tunisian revolution, when both Al Jazeera and France24 managed to circumvent state control and disseminate unrestricted information about the protests via satellite, a technology that the Tunisian government did not have the technical capacity to censor or block to the same degree as others.[15]

The technological flexibility brought about by the rise of the Internet created other additional avenues for free expression. In 2007, six radio stations, whose owners were waiting for government approval for broadcasting licenses, formed Tunisia's first Internet radio station, Radio 6. The goal of Radio 6 was to support freedom of expression and encourage access to the airwaves. "We waited a long time for licenses from the Tunisian authorities to broadcast our programming. Eventually we met and decided to unify our efforts to launch a radio station on the Internet, through which we support the freedom of expression," said Saleh Forti, then-Secretary General of the Tunisian Union of Free Radio Stations.[16] The station owners took advantage of the vague nature of Internet broadcasting regulation, which was minimal compared to the hurdles they faced in their unsuccessful attempts to obtain traditional broadcasting licenses and frequency allocations.[17]

In addition to these more subtle forms of resistance to the Ben Ali regime's control over information networks in Tunisia, direct cyber-dissent became increasingly popular—even in the face of the government's extensive and sophisticated censorship machinery and overt intimidation techniques. In the months leading up to the revolution, several key events revealed the growing cracks in the regime's authority and its increasingly frantic attempts to

[13] Committee to Protect Journalists, "10 Worst Countries to Be a Blogger," Apr. 30, 2009, accessed Dec. 13, 2013, http://cpj.org/reports/2009/04/10-worst-countries-to-be-a-blogger.php.

[14] Lucie Morillon and Jean-François Julliard, "Web 2.0 vs Control 2.0," Reporters Without Borders, Mar. 18, 2010, accessed Dec. 13, 2013, http://en.rsf.org/web-2-0-versus-control-2-0-18-03-2010,36697.

[15] Wagner, 1298 ("While modifying or even dismantling Internet capability was possible in a matter of hours, removing satellite dishes from houses across Tunisia was not an option that could be carried out easily or quickly.") For more on Al Jazeera's satellite broadcasting during the Arab Spring, see Lawrence Pintak, "The Al Jazeera Revolution," *Foreign Policy*, Feb. 8, 2011, accessed Dec. 14, 2013, http://www.foreignpolicy.com/articles/2011/02/02/the_al_jazeera_revolution#sthash.u5sTSOGo.dpbs.

[16] Jamel Arfaoui, "Tunisia Welcomes First Internet Radio Station," Magharebia, Dec. 17, 2007, accessed Dec. 14, 2013, http://www.magharebia.com/cocoon/awi/xhtml1/en_GB/features/awi/features/2007/12/17/feature-01.

[17] It is important to note that these web radio stations were hosted outside of Tunisia, because the independent Tunisian ISPs were afraid that their content might anger the Ben Ali government and did not want to be held liable for any offending content.

exert control, including the December 2010 release of a damaging collection of documents by Nawaat, a blogging platform for Tunisian dissidents, in partnership with WikiLeaks.[18] When protests against the regime in Sidi Bouzid and other cities began several days later, Ben Ali immediately ordered the ATI to maximize censorship across the country, blocking a number of sites wholesale and severely limiting access to others, including YouTube, Dailymotion, and various social networks that Tunisians had been using to share information about the demonstrations.[19] Government operators were also engaged in phishing, spoofing, and outright hacking techniques to infiltrate communications networks, obtain login credentials to social networks, and block some users' access. Independent ISPs began to throttle consumer bandwidth and cap large file transfers, possibly in an attempt to make it more difficult for users to share news and images of the protests within Tunisia and with the outside world. Users reported mobile data service outages in some of the protest locations, although it remains unclear whether these were deliberate or simply network load failures.[20]

Tension between the government and its critics reached new heights during the first week of January 2011, as groups like Anonymous, a loosely-defined coalition of web activists and hackers, announced their support for the fight against censorship.[21] On January 7, Tunisian authorities responded by arresting a great number of cyber-dissidents in the country, targeting three key groups: activists, journalists, and bloggers.[22] Ultimately, however, the government's attempt to silence its critics only further inflamed the uprising in the streets, with violent clashes erupting between protesters and police between January 8 and January 12. In a last ditch effort to win back the favor of the citizenry, Ben Ali made a televised address on January 13, announcing a number of concessions, including his plans to institute widespread reforms, remove restrictions on the Internet and free speech, and release all those who had been detained. He also revealed that he did not plan to seek reelection in 2014.[23] Within a few hours, the changes began to take effect—although the eleventh-hour overture would ultimately not be enough to save Ben Ali's presidency.

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[18] Layla Caledoniyya, "Wikileaks: The Tunisian Edition," Nawaat, Dec. 1, 2010, accessed Dec. 14, 2013, http://nawaat.org/portail/2010/12/01/wikileaks-the-tunisian-edition/. The collection of seventeen documents, which bloggers and reporters referred to as "Tunileaks," were released in December 2010 by Nawaat and Wikileaks. They included communications between the U.S. Embassy in Tunisia and the U.S. State Department from 2008 to 2010 that revealed an unflattering picture of the public's opinion of Ben Ali and other government ministers.

[19] Social network censorship began in 2007, continued with the shutdown of Facebook for ten days in August 2008, was intensified by the Presidential election of 2009, and culminated in 2010. See Riadh Ferjani, "All the Sides of Censorship: Online Media Accountability Practices in Pre-Revolutionary Tunisia," Working Paper No. 12/2011, MediaACT, June 2011, accessed Dec. 14, 2013, http://www.mediaact.eu/fileadmin/user_upload/WP4/MediaAcT_Working_Paper_Tunisia.pdf.

[20] Maher and York.

[21] Yasmine Ryan, "Tunisia's Bitter Cyberwar," Al Jazeera, Jan. 6, 2011, accessed Dec. 14, 2013, http://www.aljazeera.com/indepth/features/2011/01/20111614145839362.html.

[22] Yasmine Ryan, "Tunisia Arrests Bloggers and Rapper," Al Jazeera, Jan. 7, 2011, accessed Dec. 14, 2013, http://www.aljazeera.com/news/africa/2011/01/20111718360234492.html.

[23] Ryan Rifai, "Timeline: Tunisia's Uprising," Al Jazeera, Jan. 23, 2011, accessed Dec. 14, 2013, http://www.aljazeera.com/indepth/spotlight/tunisia/2011/01/201114142223827361.html.

## *Progress in the Post-Revolutionary Period*

President Zine el-Abidine Ben Ali's departure from Tunisia on January 14, 2011 was a pivotal moment in the country's history. His resignation opened the door for Tunisia to hold its first real democratic elections. This landmark event led to a call for a general overhaul of media policies in Tunisia, including the creation of a robust legal framework for regulating the media and for issuing licenses for new types of media.[24] Although these regulatory agencies technically existed under the previous regime, the opaque and often informal nature of those relationships had rendered the system complex and largely ineffective, particularly in supporting the government's newly prioritized democratic goals of open access and fostering a free media environment. Just as political repression in Tunisia had been linked to strict control over telecommunications and broadcasting, promoting political freedom now requires the development of a transparent regulatory system that can adequately address these challenges.

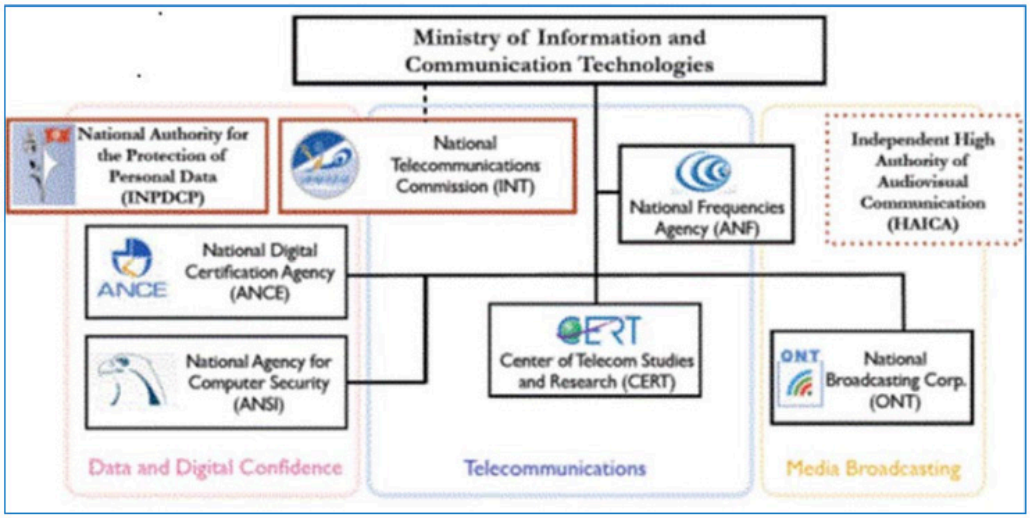



Figure 2: Regulatory agencies and authorities involved in regulation with the Ministry of Information and Communication Technologies after the Tunisian revolution.

The Ministry of Information and Communication Technologies continues to oversee most of the regulation in post-revolutionary Tunisia, although many of the pernicious practices associated with the Ministry of Information have been dismantled. The ATI, once a symbol of Ben Ali's authoritarian rule in Tunisia, has undergone dramatic changes in the name of openness and transparency, including revealing details about the censorship and surveillance it carried out during the previous regime and shifting its role towards the technical operation of Tunisia's infrastructure as an Internet Exchange Point (IXP).[25] The Agency, which previously authorized ISPs with the approval of the MoICT, no longer has any regulatory role and no authority or formal involvement in the system under the MoICT and the INT.

[24] "Reporters Without Borders in Tunisia: A New Freedom That Needs Protecting," Reporters Without Borders, Feb. 11, 2011, accessed Dec.14, 2013, http://en.rsf.org/tunisie-reporters-without-borders-in-10-02-2011,39519.html.

[25] Bryan Tan, "Censorship Authorities Will Still Have Work Post Revolution," *ZDNet*, Nov. 5, 2013, accessed Dec. 14, 2013, http://www.zdnet.com/sg/censorship-authorities-will-still-have-work-post-revolution-7000022460/.

Figure 2 above lays out the restructured regulatory system developed after the revolution, highlighting the shifts after 2011, including the removal of the ATI from the structure and the establishment of an independent authority that will be discussed in more detail below.

The latter, Decree Law No. 2011-116, outlines the creation of an Independent High Authority of Audiovisual Communication (HAICA), which will be responsible for broadcast licensing, content regulation, and public accountability.[26] Article 16 discusses the specific competencies of the agency, declaring that broadcast licensing will be regulated by HAICA in coordination with the authority on spectrum management. However, while the Decree defines the composition of the regulatory body, its powers, and its possible budgeting scheme, it is not "a substantive norm that regulates material aspects related to the provision of audiovisual media services, such as licensing, content, public service broadcasting, and so on…"[27] In essence, the Decree outlines the broad scope and structure of the agency but no specific processes. Thus, while it is an important first step—and still places Tunisia ahead of most MENA countries in liberalizing its spectrum and media regulations—it lacks the specificity to actually operationalize these new responsibilities. In particular, while it suggests possible funding, the Decree does not concretely define revenue streams, and while HAICA will likely play a role in supervising a Tunisian public broadcaster if it is established, the law itself does not include specific provisions about public service audiovisual media.[28] This lack of specificity continues to inhibit the development of a more robust and accessible licensing regulatory infrastructure, which in turn limits the flourishing of a free media environment.[29]

On April 30, 2012, INRIC released a comprehensive report that details the Committee's recommendations to the government and other major stakeholders in Tunisian public and private broadcasting and Internet services. The goal of the report was to help ensure that the Tunisian people have the right to objective, free, and pluralistic information through legislative reform. In its overview of the existing media environment in Tunisia, the report identifies significant challenges stemming from the fact that before the revolution, "Laws were either nonexistent, as in the case of the audiovisual sector, or draconian, like the press code of 1975."[30] Ultimately, the Committee argues that among necessary legislative reforms, Tunisia must establish independent regulatory bodies in the areas of broadcast, print, and electronic media "to try to bring some order and to address the highly complex problems that are posed."[31]

A number of INRIC's specific recommendations bear directly on spectrum management, broadcasting, and freedom of expression. First and foremost, the report urges the inclusion of "the principle of freedom of expression, press, information, and digital communication"

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[26] Mir, "The New Tunisian Legislative Framework: A Focus on Press and Audiovisual Media," 11.

[27] Ibid., 10.

[28] Ibid., 12–13. See also Sana Sbouai, "Média: La HAICA enfin mise en place," Nawaat, May 3, 2013, accessed Dec. 14, 2013, http://nawaat.org/portail/2013/05/03/media-la-haica-enfin-mise-en-place/.

[29] Yasmine Najjar, "Media Appointments Spark Controversy in Tunisia," Magharebia, Aug. 22, 2013, accessed Dec. 14, 2013, http://magharebia.com/en_GB/articles/awi/features/2013/08/22/feature-01.

[30] National Authority for the Reform of Information and Communication (Republic of Tunisia), "INRIC Report: Executive Summary," report, Apr. 2012, accessed Dec. 14, 2013, http://www.inric.tn/fr/INRIC-Report-Eng-final.pdf.

[31] Ibid., 19.

in Tunisia's new constitution.[32] The Committee also recommends giving constitutional status to HAICA, thereby ensuring the institution's permanence and authority, and enshrining principles against filtering of websites and electronic journals or blogs "for any reason whatsoever."[33] On the public broadcasting side, the report outlines the creation of management and governance mechanisms for public radio and television organizations. With regard to private broadcasting, it emphasizes the need to preclude the state broadcasting agency, the ONT, from "all forms of political and economic interference." INRIC also calls for an independent body to analyze the national radio frequency spectrum on the FM band to determine which frequencies are available at the local, regional, and national levels. In order to give more autonomy to the regional media, the report calls for an assessment of the characteristics and needs of each region and its respective audience, and urges the reduction of broadcasting fees for new regional radio stations.[34] Finally, on the issue of the role of the ATI and the oversight of Internet services in Tunisia, the Committee notes the need to "define the relationship between the various stakeholders," including telecom operators and ISPs, and to review the tasks of the ATI (as an Internet Exchange Point) and other bodies that monitor and regulate the Internet.[35]

## *Ongoing Challenges in Tunisia*

Instead of completely eliminating and replacing the legal institutions and structures that existed under the previous regime, government leaders and independent stakeholders in Tunisia have sought to modify and reshape these organizations to better adhere to democratic principles and values demanded by the populace. There is merit to the strategy of reforming existing structures instead of creating new ones, but it is also challenging, as it requires patient, diligent work to shift the systems, interests, and public perceptions that have existed for years. Many of these institutions continue to operate under the same people, rules, and regulations as under the previous regime. As Mir writes, "In institutional terms… Tunisia still does not have an administrative and judiciary structure ready to apply democratic laws and to perform according to non-authoritarian principles. The core of the executive power and public administration remains roughly the same."[36] The continuing political instability in Tunisia has only amplified this problem and poses a challenge to comprehensive regulatory reform.[37]

The INRIC report spells out a number of positive recommendations. However, INRIC's role as an outside entity, established after the revolution, proved to be both a strength and a weakness; while the Committee's independence gave it significant influence and power to confer legitimacy on the Tunisian reform process, it also provided it with little room to actively make the changes it recommended. In July 2012, after the completion of its review,

[32] Ibid., 21.

[33] Ibid.

[34] Ibid., 24.

[35] Ibid., 26.

[36] Mir, "The New Tunisian Legislative Framework: A Focus on Press and Audiovisual Media," 14.

[37] "Tunisia Head-to-Head: What Next?" BBC News, Aug. 27, 2013, accessed Dec. 14, 2013, http://www.bbc.co.uk/news/world-africa-23790897.

Laabidi announced that INRIC would shut down, expressing some doubts about the future of the media environment in Tunisia.[38] The report itself echoes similar concerns that the government may not actually implement many of these reforms, particularly as it has not yet taken concrete steps to establish and fund HAICA, even though the law was passed in November 2011.[39] In that vein, while the state broadcaster is slowly adapting to the new, post-revolutionary culture in Tunisia (despite having previously served as a virtual propaganda arm of the government), it has not yet established itself as acting reliably in the public interest.[40]

Significant barriers to private entry into the broadcasting sector remain as well. The ONT retains exclusive rights to own broadcasting transmitters, and rents them to private broadcasters for an additional fee beyond the license for spectrum assignment by the ANF. In total, these agreements incur fees of about $60,000 USD (110,000 TND) for most of the new radio stations, such as OxygenFM, IFM, and KalimaFM. For many broadcasters, especially local and independent media groups, these costs are prohibitively expensive. Consequently, despite recent changes in the media environment, a number of broadcasters have been forced to continue to use the airwaves illegally. The proliferation of FM pirate radio stations operating illegally in post-revolutionary Tunisia, as well as the growth of online radio stations, is evidence of the need for greater access to the airwaves for new market entrants, and a tendering and licensing system that enables such access.[41] Pirate radio highlights the need for much more extensive reforms of the system that will create legal opportunities for a truly diverse range of broadcasters.

The final act of Ben Ali's presidency—lifting the Tunisian censorship regime—represented an important first step toward the creation of a free media environment in Tunisia. While this act has been celebrated by many and was reflected in Tunisia's removal from the Reporters Without Borders "Enemies of the Internet" list, it has not yet been accompanied by a corresponding effort to explicitly codify rights to freedom of expression and protection from censorship in the laws that govern wired and wireless access to the Internet. Moreover, while the government has discussed—and recommends in the INRIC report—recognizing Internet freedom as a global principle and enshrining the right to access information, it has only begun to take initial steps to enact these reforms. On September 4, 2012, the government announced the launch of a National Forum on Internet Governance, which Information and Communications Minister Mongi Marzoug declared to be the end of Internet censorship in Tunisia.[42] A few days later, Tunisia joined the Freedom Online Coalition[43]—"a global dialogue about the responsibilities of governments from around the world in pro-actively furthering

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[38] "Tunisia Media Authority Shuts Down, Citing Censorship," Al Arabiya News, July 4, 2012, accessed Dec. 14, 2013, http://english.alarabiya.net/articles/2012/07/04/224438.html.

[39] "Tunisia: NAICR Calls for Enshrining Audiovisual Regulation Authority in New Constitution," Tunique Afrique Presse, June 13, 2012, accessed Dec. 14, 2013, http://allafrica.com/stories/201206140458.html.

[40] Mir, "The New Tunisian Legislative Framework: A Focus on Press and Audiovisual Media," 13.

[41] Ghislain Fornier de Violet, "Radios: les pirates à l'assaut des ondes," Medinapart, Mar. 19, 2012, accessed Dec. 14, 2013, http://www.medinapart.com/2012/03/19/radios-les-pirates-a-lassaut-des-ondes/.

[42] "Tunisia: Internet and Communication Minister—'Internet Censorship No Longer Implemented in Tunisia'," Tunique Afrique Presse, Sept. 6, 2012, accessed Dec. 14, 2013, http://allafrica.com/stories/201209070049.html.

[43] Freedom Online Tunis, "Tunisia Joined the Freedom Online Coalition: Lessons Learned!" accessed Nov. 14, 2013, http://www.freedomonline.tn/Fr/tunisia-joined-the-freedom-online-coalition-lessons-learned_11_51.

freedom on the internet"—and hosted the Coalition's third meeting in Tunis in June 2013.[44] It remains to be seen how these developments will impact the legal and regulatory framework in practice in Tunisia in the long term.

In terms of infrastructure and access, growth has continued while citizens have largely enjoyed unfettered access to the global Internet since the revolution. In December 2011, the Ministry of Information and Communication Technologies reported 12.5 million mobile subscriptions, with a mobile penetration rate of 116.6%, and an estimated 4.6 million Internet users, with broadband Internet penetration at around 9%.[45] In large measure, these users have been able connect to the global network unencumbered, particularly via social networks, which were heavily regulated before the revolution. However, the Tunisian Internet Agency continued to face filtering demands for months following Ben Ali's departure. In May 2011, the Military Tribunal requested the filtering of five Facebook pages, with which the ATI originally complied.[46] But the rise in Internet traffic in the post-revolutionary period created a technical dilemma for the ATI: with the existing machinery, the agency would have to significantly throttle the speed of Internet traffic in order to block the pages. Shortly thereafter, a civil lawsuit requesting that the ATI block certain pornographic websites was also filed. There are ongoing legal inquiries into whether the ATI will have to comply with requests of this nature going forward, even as it has attempted to remain neutral as an Internet exchange point and to enable Internet freedom for its citizens.[47] Moreover, the quality of connectivity and access also varies greatly from region to region, largely because Tunisia lacks an adequate universal service policy and maintains an unclear regulatory framework regarding the quality of service obligations on existing ISPs as well as fixed and mobile operators.

## *Recommendations for Spectrum Regulation and Reform in Tunisia*

The public commitment to online freedom in Tunisia and to the ideals of freedom of the media and freedom of expression in the post-revolutionary period has been encouraging. However, fostering a truly democratic media environment requires a much broader set of policy reforms beyond Internet regulation. In particular, spectrum policy can be a powerful tool for enabling access to information, freedom of expression, and freedom of the media, and as such can and must be a part of the conversation as Tunisia seeks to overhaul its communications policies to deliver on the demands of its recent revolution. Moreover, this post-revolutionary environment remains a unique opportunity to explore different regulatory mechanisms, which could serve as a model particularly to similarly-situated countries in

[44] Freedom Online Tunis, "About the Freedom Online Coalition," accessed Nov. 14, 2013, http://freedomonline.tn/Fr/about-the-freedom-online-coalition_11_50.

[45] Data compiled on Nov. 14, 2013 from MinCom (Ministry of Information and Communication Technologies, Republic of Tunisia), "Indicators," http://www.mincom.tn/index.php?id=361&L=2. The mobile penetration rate statistic is calculated as number of mobile subscriptions per 100 inhabitants. Internet broadband penetration is calculated as number of broadband internet subscriptions per 100 inhabitants.

[46] Tunisian Internet Agency, "List of Filtering Requests," accessed Aug. 23, 2012, http://filtrage.ati.tn/. One of the present authors, Moez Chakchouk, is the head of this agency.

[47] Jillian C. York, "Fighting Censorship on Principle," Al Jazeera, Feb. 23, 2012, accessed Dec.14, 2013, http://www.aljazeera.com/indepth/opinion/2012/02/201222381652993947.html; "Tunisia's Highest Court Overturns Ruling on Filtering of Pornography Sites," Reporters Without Borders, Feb. 22, 2012, accessed Dec. 14, 2013, http://en.rsf.org/tunisia-internet-filtering-danger-of-03-02-2012,41805.html.

the region, but also more broadly. In this section we seek to explore potential opportunities for spectrum policy reform which would ensure a freer media environment, and would be part of a shift to better reflect Tunisia's recent commitments to Internet freedom in practice. While the policies discussed in this section are complementary and could be implemented together, they are discussed here to highlight the different policy tools and considerations that the Tunisian regulatory authorities have at their disposal.

There are traditionally three regulatory approaches to spectrum management. Historically, most telecommunications have been governed under the "command and control" method wherein airwaves were considered to be a public good.[48] Tunisia's telecommunications sector was largely managed under this structure until the late 2000s, when the government began to liberalize its policies and adopt a modified market-based "property rights" approach,[49] which treats spectrum as a resource that can be bought, sold, and often traded on the open market.[50] More recently, certain policymakers and academics have advocated for an open access "commons" approach, which in its most radical incarnation posits that the customary licensing regime is unnecessary as technology has advanced to such a degree that devices can share spectrum without interference.[51]

Interestingly, those that favor a commons-based approach and those that favor property rights largely utilize the same language—those who support the latter argue that open access would lead to a "tragedy of the commons," whereas proponents of the commons approach worry about the "tragedy of the anti-commons."[52] Both sides argue that their management style preserves democratic freedoms and free market ideals.[53] Many Western countries, including the United States, Canada, and the United Kingdom, have adopted a hybrid approach to regulation, maintaining aspects of command and control while heavily investing in market-based solutions and enabling open access experimentation.[54] Tunisia and other MENA countries undergoing radical shifts in governance structures are uniquely situated to benefit from the lessons provided by these countries. The creation of a regulatory framework that can evolve quickly in accordance with technological and social changes is critical for the continued democratic and commercial growth of the Tunisian nation. Flexibility is particularly important in this enterprise. Every year, the amount of spectrum required to transmit data packets decreases; according to wireless pioneer Martin Cooper, spectral efficiency doubles every 30 months.[55] Even so, given the explosive growth in mobile subscriptions and broadband Internet access in Tunisia over the past decade, a new approach will be

[48] Benoît Freyens, "A Policy Spectrum for Spectrum Economics," *Information Economics and Policy* 21 (2009): 128.

[49] Cankorel and Aryani.

[50] Ronald H. Coase, "The Federal Communications Commission," *Journal of Law and Economics* 2 (1959): 32.

[51] Kevin Werbach, "Supercommons: Toward a Unified Theory of Wireless Communications," *Texas Law Review* 82 (2004): 865–866.

[52] Kevin Werbach, "The Wasteland: Anticommons, White Spaces, and the Fallacy of Spectrum," *Arizona Law Review* 53 (2011): 217–218.

[53] Jeffrey L. Blevins and Sarah C. Barrow, "The Political Economy of Free Speech and Network Neutrality: A Critical Analysis," Journal of Media Law & Ethics 1 (2009): 27–48.

[54] Freyens, 128.

[55] Steve Song, "Spectrum for Development: The Impact and the Role of Wireless Technologies," white paper, Association for Progressive Communications (2011), accessed Dec. 14, 2013, http://www.apc.org/en/system/files/OpenSpectrumIssuePaper_EN.pdf.

necessary if Tunisia's regulators have any hope of keeping pace with growth and innovation going forward, maintaining the country's competitive edge in the region, and realizing the opportunities created by the revolution.

There are a number of key ways that spectrum policy in Tunisia can be redesigned to enable democratic access, facilitate freedom of expression, and encourage economic growth. In this section we outline recommendations for Tunisian spectrum regulation. These recommendations are meant to balance the interests of commercial actors and the public interest, and enable policies for technological innovation. We emphasize the need for the Tunisian government to construct these regulations in a participatory and democratic manner, building a legacy of transparency and accountability that counters the memory of the previous regime. Specifically, we touch on broadcasting licenses, wireless access, and open spectrum use.

## Broadcast Licensing

One of the key decisions facing the Tunisian government is how to structure the licensing regime for radio and TV broadcasting so that it fosters the democratic spirit of the transition while encouraging economic growth. This balance may be difficult because, as has been mentioned previously, the line between the former regime and the incumbent broadcasting and telecommunications operators in Tunisia is blurry at best; while Ben Ali and his family have been forced out of the country, many of the allies of the previous government have retained a great deal of power and influence today. Currently, anyone hoping to obtain a new license or to legally broadcast in a vibrant and free media environment in Tunisia will find themselves hindered by a convoluted and highly subjective regulatory process.

However, there are steps the government can take to eliminate these problems. In line with the goal of increased transparency, the first step should be to provide the public with information about how different bands are designated for different purposes, which frequencies have been assigned and whether they are currently in use, and how to delineate the process for licensing and allocation. As the INRIC report suggests, this process will likely require a survey of the available frequencies. Once that information has been compiled and the licensing procedures have been defined, the logical next step in the process of democratic reform will be to give the public the opportunity to offer input on the licensing process, and to make available the justification for subsequent decisions in order to establish a precedent for implementing the criteria. Any further restructuring should be as simple and transparent as possible, with clearly defined objective criteria for licensing.

We submit that there are three key elements for the Tunisian government to consider while designing the broadcast licensing regime: plurality, development, and democratic participation. These elements would help create a broadcasting ecosystem that is diverse and accessible to a wide range of actors, and which relies on public and private organizations that are both commercial and non-profit. While the focus has often been on national and regional broadcasting, we also argue for the importance of promoting and protecting community broadcasting in Tunisia. As such, there are several important matters to emphasize in policy development.

**Create opportunities for different types of broadcast media outlets, including community broadcasting:** The license fee structure should take into account both the needs and financial abilities of different categories of broadcasters. TV and radio operators who intend

to broadcast nationwide should be regulated separately from community radio stations and television channels, and the ONT should strive to balance the differences between the two in their approach to licensing. The authors of Broadcasting, Voice, and Accountability outline a number of key requirements when attempting to create an enabling environment for community broadcasting.[56] First, community broadcasting should be recognized by law as a distinct type of broadcasting.[57] "Community broadcasting" can be defined as independent broadcasting that is provided by and for the members of a community in a particular geographical location or that belongs to a particular community of interest. Its primary purpose is to deliver social benefit and not to operate for private commercial profit.[58] It should be owned by and accountable to the community that it seeks to serve and it should provide for participation by the community at all levels.[59]

The European Parliament has affirmed that community media constitute a "significant component of participatory democracy."[60] The study commissioned by the Parliament concluded that community media fulfill local informational needs; enable citizens to play an active role in community life; act as a catalyst of cultural and artistic initiatives, local public interest activities and civil society advocacy; and encourage the direct involvement of typically underrepresented groups such as ethnic minorities and young people.[61]

Community broadcasting should be supported and encouraged by spectrum regulators through specific and explicit licensing arrangements which guarantee fair and equitable access to radio spectrum for civil society and community-based organizations. The license terms and conditions for community broadcasters, while consistent with the overall objectives of broadcast regulation, should also be designed to ensure that community broadcasting service characteristics are preserved for the duration of the license period. Furthermore, to ensure independence, community broadcasting should be supported by public funding and the funding processes should be determined and administered by an independent public body that is objective, transparent, and accessible to the public at large.[62]

**Allow licenses for low power FM broadcasting:** As part of its support for community broadcasting, the spectrum regulator could also offer licenses for low power FM (LPFM) broadcasting, following the examples of other countries that have implemented systems to encourage the creation of local, community content. In 2000, the U.S. Federal Communications Commission created the Lower Power FM Radio Service, which authorizes non-commercial, educational broadcasters to operate on FM bands at an effective radio power of up to 100

[56] Steve Buckley, Kreszentia Duer, Toby Mendel, and Seán Ó Siochrú, *Broadcasting, Voice, and Accountability: A Public Interest Approach to Policy, Law, and Regulation* (Ann Arbor, MI: University of Michigan Press, 2008).

[57] Ibid., 156–160.

[58] A study commissioned by the European Parliament further defines community media as: "media that are non-profit and owned by or accountable to the community that they seek to serve. [They] are open to participation in program making and management by members of the community." European Parliament, "The State of Community Media in the European Union," report, Sept. 2007, accessed Dec. 13, 2013, http://www.cmfe.eu/docs/1166366638_The%20state%20of%20community%20media.pdf, iii.

[59] Kate Coyer and Arne Hintz, "Developing the 'Third Sector': Community Media Policies in Europe," in *Media Freedom and Pluralism: Media Policy Challenges in the Enlarged Europe*, ed. Beata Klimkiewicz (Budapest: Central European University Press, 2010), 275–298.

[60] European Parliament, 5.

[61] Ibid., 51–55.

[62] Buckley, Duer, Mendel, and Siochrú.

watts.[63] Applicants must pay filing and administrative fees, but if they are approved, the licenses themselves are provided free of charge. Their stations must meet certain environmental and equipment standards and agree to accept any interference on their channels from other licensed FM operators, relying on resources like the Low Power FM Channel Finder to identify available FM broadcast channels in their communities.[64]

Because of its extremely local nature (in the U.S. the standard range for an LPFM station is about 3.5 miles or 5.6 kilometers[65]), LPFM broadcasting is ideal for community broadcasters providing educational and cultural content, as well as transportation and public safety organizations. Allowing these broadcasters to share certain spectrum bands will help promote a free media environment and encourage local content targeted at a specific audience.

## Wireless Communications

In addition to broadcasting, spectrum has the power to facilitate greater access to the Internet through the use of wireless technology. This mechanism for access is particularly important in some of the least-connected rural communities, which are isolated from coastal hubs by both physical distance and lack of virtual connectivity to information networks.[66] Recent studies have found that Tunisians in the interior provinces of the country are far less likely to use mobile phones, connect to social networks, or even have access to computers and the Internet than their peers in coastal areas. For example, a comparison of connectivity shows that while 85.8% of the population in the coastal Nabeul province are mobile subscribers, that figure is only 63.8% in the inland province of Kasserine. There are a total of 27 public Internet access points in all of Tunisia's inland provinces, compared to 232 in the coastal provinces.[67] The rural broadband gap points not only to economic inequality in Tunisia, but also to the comparative lack of infrastructure. This disparity in infrastructure is underscored by the fact that obtaining fixed-line access to high-speed Internet is comparatively easy in Tunisia's coastal cities, contributing to the already high rates of broadband penetration nationally. By contrast, the rural interior of the country will likely be opened up by the proliferation of wireless technologies, including mobile and fixed wireless Internet. Although wireless connectivity may only be a temporary measure as Tunisia develops the capacity to provide high quality fiber access to more and more areas of the country, it is a critical step that can help stem the growing divide between the interior and coastal areas of the country. Thus, Tunisian spectrum policy should be designed to specifically address the issue of universal access. To that end, we offer the following suggestions:

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[63] Federal Communications Commission (United States), "Low Power FM Broadcast Radio Stations," accessed Dec. 14, 2013, http://www.fcc.gov/encyclopedia/low-power-fm-broadcast-radio-stations-lpfm.

[64] Federal Communications Commission (United States), "Low Power FM Channel Finder," accessed Dec. 14, 2013, http://www.fcc.gov/encyclopedia/low-power-fm-lpfm-channel-finder.

[65] Federal Communications Commission (United States), "Low Power FM Broadcast Radio Stations."

[66] World Bank, "Poor Places, Thriving People: How the Middle East and North Africa Can Rise above Spatial Disparities," MENA Development Report (2010), accessed Dec. 14, 2013, http://siteresources.worldbank.org/INTMENA/Resources/OverviewENprintersversionspatialdisparities.pdf, 3–10.

[67] Zack Brisson and Kate Krontiris, "Tunisia: From Revolutions to Institutions," white paper, InfoDev and the World Bank (2012), accessed Dec. 14, 2013, http://www.infodev.org/infodev-files/resource/InfodevDocuments_1141.pdf.

**Exercise caution about spectrum auctions: Spectrum auctions have been utilized in many Western countries and are a hallmark of the "property rights" governance structure**.[68] This approach may be extraordinarily tempting to developing nations because it can raise substantial sums of money for the government in a short time. These auctions should be considered with caution in Tunisia and only implemented with certain caveats. One of the biggest risks associated with the auction system is that large companies often have both the incentive and the financial resources to buy as much available spectrum as possible. In order to prevent incumbent providers from purchasing all of the prime spectrum at auction, any regulation regarding the practice of auctions should stipulate that certain bands can be sold only to smaller companies and non-profits, thereby ensuring that auctions are not simply an opportunity for incumbent providers to muscle out any competition.

In general, the auctions currently used to allocate spectrum licenses tend to favor operators with considerable market power, and this rarely leads to lower prices for the consumer.[69] Although these auctions have been successful by certain standards, they have also led to a highly concentrated market structure. The auction model encourages bidding from companies that intend to maximize profits from the spectrum they purchase rather than create competitive pressure that would lead to lower prices across the board for consumers. "Put another way, auction theory tells us that the type of auction mechanisms used to allocate spectrum, if they work well, will tend to maximize industry profits, and one expects industry profits to be higher with more concentration," write Milgrom, Levin, and Eilat.[70] A number of scholars, including Nobel Laureate Elinor Ostrom, have proposed alternative governance systems for managing "common pool" resources like spectrum. She outlines a number of principles that enable this type of management.[71]

**Limit inactivity:** It is also necessary to be wary of "spectrum squatting," a process that allows companies to buy up available spectrum as an anticompetitive measure and then leave it unused to create artificial scarcity. By limiting the amount of time that a company can maintain a license for fallow spectrum bands, the government can free up valuable spectrum for legitimate purposes. Similar measures have been adopted in Australia and elsewhere to positive effect.[72] In addition to encouraging a competitive market, these measures will prevent allies of the former regime from maintaining media control through exclusive spectrum licenses.

**Permit license-exempt wireless use:** To complement better policies governing the use of exclusively licensed spectrum, it is also critical that Tunisia allow and even encourage license-exempt wireless use. Examples from other nations show that it is often entrepreneurs operating on license- exempt spectrum and using widely available Wi-Fi equipment that provide the middle- and last-mile connectivity for communities without access to wireline

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[68] Eli Noam, "Spectrum Auctions: Yesterday's Heresy, Today's Orthodoxy, Tomorrow's Anachronism. Taking the Next Step to Open Spectrum Access," *Journal of Law and Economics* 41 (1998): 765–790.

[69] Paul Milgrom, Jonathan Levin, and Assaf Eilat, "The Case for Unlicensed Spectrum," SIEPR Discussion Paper No. 10-036, Stanford Institute for Economic Policy Research, Oct. 12, 2011, accessed Dec. 14, 2013, http://siepr.stanford.edu/?q=/system/files/shared/pubs/papers/pdf/10-036_Paper_Milgrom.pdf.

[70] Ibid., 12–13.

[71] Elinor Ostrom, *Governing the Commons: The Evolution of Institutions for Collective Action* (New York: Cambridge University Press, 1990).

[72] Martin Cave, "Anti-Competitive Behaviour in Spectrum Markets: Analysis and Response," *Telecommunications Policy* 34 (2010): 251–261.

infrastructure. In the United States, for example, wireless Internet service providers (WISPs) have been integral in providing access to rural pockets of the country, primarily in areas where cable and telecom companies have avoided building the infrastructure for wired Internet service. Currently, there are over 2,000 WISPs providing service to approximately two million Americans, many of whom live in areas that would not otherwise have high-speed Internet access—or any access at all.[73] The opportunity for similar deployments exists in Tunisia today, as there is growing demand in rural areas to harness the economic potential that corresponds with increased Internet access, as well as a burgeoning desire among the country's residents to participate in the expanding community of online activism and political debate.

The relative ease of establishing robust connectivity in rural Tunisia using wireless technology was demonstrated poignantly in the refugee camps on the border with Libya during the Libyan refugee crisis of 2011. The Tunisian Red Crescent and the International Committee of the Red Cross set up makeshift camps to accommodate the large numbers of Libyans flowing across the border to escape the violence in their country, and Telecoms Sans Frontiers *(TSF)*, a non-governmental organization, assisted by providing emergency telecommunications relief, offering satellite phones for people to call loved ones, and establishing a satellite Internet connection for emergency workers in the camps. Using Wi-Fi routers, TSF was able to distribute quality connectivity throughout the camps, in an area of the country where Internet connections are normally few and far between and Wi-Fi hotspots typically require a license to operate.[74] While these connections were established ad hoc during an emergency situation, they highlight a real opportunity to provide permanent wireless connectivity in some of Tunisia's least-connected regions using similar technology.

It is even possible to go a step further if the regulatory agency allows for different power requirements on wireless equipment depending on the region. Many countries set a universal power limit on all Wi-Fi devices to prevent harmful interference. While this approach may be necessary in densely populated urban areas where the 2.4 and 5 GHz bands operate near or at capacity, it is far less of a concern in rural areas. Regional power requirements that allow far more flexibility in sparsely populated areas would facilitate greater wireless connectivity without having to lay more fixed-line infrastructure in those areas.

**Utilize a portion of the digital dividend for rural development:** Along these lines, the digital switchover, which is scheduled to take place in the MENA region in 2015, could further increase the amount of spectrum available for development in rural areas. As television broadcasting shifts from analog to digital, it will free up valuable spectrum on the lower end of the frequency range (often referred to as "white space") that enables wireless signals to travel further and cover greater distances. Rather than simply repurposing this spectrum and auctioning it off, the Tunisian government has the opportunity to use the digital dividend in the interest of the public good and invest in the long term growth and development of cur-

[73] Matt Larsen, "America's Broadband Heroes: Fixed Wireless Broadband Providers, Delivering Broadband to Unserved and Underserved Americans," white paper, Wireless Internet Service Providers Association (2011), accessed Dec. 14, 2013, http://www.wirelesscowboys.com/wp-content/uploads/2011/10/americas-broadband-heroes-fixed-wireless-2011.pdf.

[74] "TSF Being Deployed along the Tunisia-Libya Border for Refugees," Wireless Federation, Mar. 4, 2011, accessed Dec. 14, 2013, http://wirelessfederation.com/news/36830-tsf-being-deployed-along-the-tunisia-libya-border-for-refugees/.

rently underserved areas of the country.[75] These lower-frequency bands would be beneficial if opened to small wireless Internet service providers on a license-exempt basis, facilitating expansion of access on frequencies that have better propagation characteristics than those traditionally reserved for Wi-Fi devices.

**Embrace the potential of open spectrum and experimentation:** Even if spectrum is heavily regulated for both commercial and public interest purposes, it is also critical that some bands of spectrum are reserved for creative uses and experimentation that can enable technological innovation. Wi-Fi is one of the most commonly cited examples of a revolutionary technology that was developed on the so-called "junk" band under a commons regulatory approach. In the 1980s, the United States made that portion of the spectrum available to anyone who complied with established technical requirements, which include power limits, equipment standards, and certification that mitigate potential harmful interference.[76] By lowering barriers to access, the commons approach drove the development of spectrum-efficient radio technologies including Ultra Wideband *(UWB)*, software-defined radios, smart radios and antennas, cognitive radios, and mesh networks.[77]

There are inherent risks involved in this approach—some say it can potentially lead to overuse of spectrum and excessive interference.[78] However, opportunistic access models *(which allow devices to operate on bands for which usage is low or intermittent, such as those reserved by governments for emergency use)* can work by relying on databases that track the frequencies that are in use and other means to allow devices to utilize spectrum that would otherwise lay fallow. The technology will continue to improve with further experimentation and use, but it will need regulators to provide opportunities for it to flourish.[79] At the same time, further innovation can be encouraged and enabled by shortening the time period for exclusive spectrum licenses, which will allow the government to shift spectrum policy more easily to take advantage of innovation. A movement away from long-term ownership or control models for spectrum and toward more temporary and flexible authorizations can ultimately lead to more efficient use of spectrum and further development of technologies that can mitigate the risks associated with open spectrum and sharing.

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[75] The value of unlicensed spectrum, not only as a mechanism to encourage wireless innovation but also as an economically valuable resource unto itself, has been shown by a number of experts. In particular, Mark Cooper makes the case for unlicensed spectrum as a far more sustainable means for growth, arguing that "focusing on short term revenues that would be raised by auctioning licenses, while ignoring the immense value that unlicensed spectrum creates, which would increase revenues much more than auction yields, is a mistake… [F]ailing to make more high-quality spectrum available for unlicensed use would severely constrain the development of the unlicensed use model, which would retard the development of the entire wireless broadband sector, stifle innovation, drive up consumer costs for service, reduce long-term growth and slow job creation." Mark Cooper, "Efficiency Gains and Consumer Benefits of Unlicensed Access to the Public Airwaves: The Dramatic Success of Combining Market Principles and Shared Access," working paper, Jan. 2012, accessed Dec. 14, 2013, http://www.markcooperresearch.com/SharedSpectrumAnalysis.pdf, 5.

[76] Julius Genachowski, "Remarks on Preserving Internet Freedom and Openness," speech, Dec. 1, 2010, accessed Dec. 14, 2013, http://www.fcc.gov/document/federal-communications-commission-chairman-julius-genachowski-remarks-preserving-internet-f.

[77] Organisation for Economic Co-operation and Development, "The Spectrum Dividend: Spectrum Management Issues," OECD Digital Economy Papers No. 125 (2007): 16.

[78] Philip J. Weiser and Dale N. Hatfield, "Policing the Spectrum Commons," *Fordham Law Review* 74 (2005): 664.

[79] Kevin Werbach, "Radio Revolution: The Coming of Age of Unlicensed Wireless," white paper, New America Foundation and Public Knowledge (2003), accessed Dec. 14, 2013, http://werbach.com/docs/RadioRevolution.pdf, 14.

### *Beyond Licensing and Allocation: Creation of an Independent Spectrum Regulator*

While the above recommendations are important, licensing and allocation alone will not suffice. Tunisian regulatory authorities must develop a system for policing unauthorized use of spectrum—from detecting, locating, and mitigating the problems created by interference to regulating content abuse. In contrast to many of the closed agencies that existed prior to the revolution, the Tunisian government should always strive for transparency and accountability in the way these issues are handled going forward. To that end, in addition to a clearly-defined and easy-to-navigate regulatory process, the government should create a viable mechanism for receiving feedback and mediating complaints. It should also provide a searchable database or registrar of license holders, so that members of the public can easily ascertain who has the rights to which spectrum bands in any given area. If civil society groups or other dedicated organizations commit to lobbying on behalf of smaller spectrum users and the public interest, the regulatory entity must be willing to listen to their concerns and include such organizations in any consultative processes.

To achieve many of the goals laid out in this article, Tunisia needs a comprehensive spectrum regulator who can clarify the roles of the various organizations and create a system that operates efficaciously. A specific agency tasked with identifying strategic convergent approaches to spectrum issues will help in balancing both local and national broadcasting and telecommunications needs, and can broaden the focus from improving Internet access in coastal cities, where fixed-line infrastructure is relatively easy to build, to including the rural interior of the country. A single spectrum regulator could also advocate for prioritizing spectrum policy in Tunisia's ICT regulatory structure and ensuring that frequencies are allocated in a manner that promotes economic growth, democratic access, and innovation. However, as Tunisia has already learned, in order to promote a democratic and free media environment, it is critical that this regulator be independent from the government.

## *Conclusion*

Tunisia's influential role in the political transformations of the Arab Spring does not have to be limited to the events of 2010 and 2011. Regional analysis suggests that Tunisia has the power to continue to lead its global peers as it completes the process of constitutional reform and establishes itself as a true democracy. The technologies that played such a critical role in the revolution will undoubtedly continue to be influential going forward, and new forms of media and online civic activities will likely continue to play an important role in shaping Tunisian political and social development. As such, it is necessary for Tunisia to preserve the most valuable achievement of the revolution—freedom of expression. With the participation of all stakeholders, Tunisia should move to carry out the necessary reforms in accordance with this principle and craft a neutral and fair regulatory framework that is favorable to economic development.

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